Supplementary material

TSHR and IGF1R in Graves disease and thyroid eye disease: genetic evidence from blood expression

Supplementary methods

Dataset definitions and analysis chronology

BBJ GD (GCST90018627) and UKB self-reported hyperthyroidism/thyrotoxicosis (GCST90038636) retained the source sample sizes and estimates. UKB used BOLT-LMM linear-model statistics; its report included adjustment for sex, age, BMI, assessment centre, ethnicity, genotyping batch and 20 principal components. The available documentation did not establish an exclusively European analysis sample. FinnGen R12 GWAS counts were 858/499,490 for GRAVES_OPHT, 3,962/496,386 for E4_GRAVES_STRICT and 753/499,595 for E4_GRAVES_OPHT_STRICT. Counts refer to the analyzed GWAS files, not later phenotype-browser totals. Strict GD combines specified GD/ophthalmopathy diagnostic criteria with the Graves condition and exclusion of amiodarone exposure; strict ophthalmopathy requires its specified ophthalmopathy and thyrotoxicosis codes. The exact endpoint definitions and source manifests are retained with the analysis verification records. None supplies GD-only controls.

The original three-outcome analysis preceded the two FinnGen extensions. The expanded analysis plan was recorded before the new endpoint associations were computed, but it was not a prospective registration of the complete study. The 24 colocalization genes comprise 13 BBJ discoveries, *IGF1R*, nine additional primary FinnGen GD discoveries, and frequency-sensitive *HLA-G*. No new discovery threshold was optimized from these results.

Instrument selection and effect reconstruction

Significant cis-eQTL instruments ($P < 5\times10^{-8}$ within ±1 Mb of the gene) were clumped at $r^2 < 0.001$ within 10 Mb using the European 1000 Genomes Phase 3 reference (n = 503). Selected instruments and GRCh37 positions are in the Instruments worksheet. $SE = 1/\sqrt{[2f(1-f)(N + Z^2)]}$ and $\beta = Z \times SE$, where f is the assessed-allele frequency and N is the variant-specific expression sample size. Unit expression variance was assumed. Variant $F = Z^2$. The estimate is approximately per expression SD, not per SD of the genetic predictor.

TwoSampleMR harmonise_data(action = 2) aligned alleles and excluded ambiguous palindromic variants with MAF > 0.42. Missing outcome estimates were not imputed. BBJ and FinnGen retained log-odds scales; UKB β and SE were each divided once by $\mu(1-\mu)$, with $\mu = 3{,}731/484{,}598$. GRCh37 expression and GRCh38 FinnGen coordinates were not directly joined: identifiers and allele pairs

were checked, with duplicate and alias audits. Source covariate adjustments were retained; patient-level re-adjustment and subgroup analysis were unavailable.

MR estimators and instrument diagnostics

For a single instrument, the Wald coefficient is β_Y/β_X and its first-order SE is $SE_Y/|\beta_X|$. Multiple-instrument MR used inverse-variance weighted regression through the origin. Multiplicative random-effects SEs included under-dispersion correction, so heterogeneity could widen the interval. Normal two-sided tests and 95% CIs were used for the primary estimators. The alternative-estimator worksheet preserves weighted median, weighted mode and MR-Egger results for the original three outcomes; mode and Egger intervals use the corresponding t distributions. These alternative estimators were not rerun for all new endpoints.

Post hoc leave-one-out analyses applied IVW when at least two variants remained and the Wald ratio when one remained. Single-instrument *TSHR* and BBJ *CTLA4* were ineligible. Original three-outcome results, new-endpoint results for the fixed 14-gene set, and 72 *C4B*/*HLA-G* omission estimates under two frequency sources are retained in separately named worksheets. They reuse the same association data and are not independent replications. *C4B*/*HLA-G* instrument weights use β_X^2/SE_Y^2, normalized within each full model. A single-SNP *P* value can be much smaller than the full IVW *P* value when heterogeneity inflates the full-model SE.

TSHR clumping sensitivity used eQTL thresholds of $5\times10^{-8}/5\times10^{-6}$, r^2 thresholds of 0.001/0.01/0.1, and European (n = 503) or East Asian (n = 504) reference panels. Primary settings selected one European and two East Asian variants, and rs179252 was retained in every setting. Chromosome 16p11.2 analyses found high East Asian LD among rs4889606, rs34649473 and rs78924645; conditional analysis supported regional dependence rather than three independent gene effects.

TSHR allele alignment and reference LD

In a post hoc audit, original full eQTLGen and disease GWAS files were re-extracted for rs179252 and rs179247. Their GRCh37 positions were 14:81435985 and 14:81432546, and GRCh38 positions were 14:80969641 and 14:80966202. Both non-palindromic allele pairs matched across sources. UKB effects were reversed to the G allele after the existing scale conversion; BBJ and FinnGen required no reversal. Exposure coefficients and first-order Wald estimates were independently reconstructed. Original GWAS *P* values and recalculated MR *P* values were retained separately.

G-allele counts at both variants were positively correlated in European (n = 503; r = 0.978322812; r^2 = 0.957115524) and East Asian (n = 504; r = 0.995803919; r^2 = 0.991625445) references, without missing genotypes. Direct BED decoding agreed with PLINK dosage output. PLINK haplotype estimates supported G–G and T–A coupling; these were inferred from unphased genotypes, not

directly observed phased haplotypes. Genotype-based r^2 and haplotype-based estimates were kept distinct. Alignment to the G allele resolved the negative default European correlation obtained when counting T at rs179252. The TSHR_allele_audit worksheet retains both eQTL records, both variants across five disease outcomes, all four inferred haplotypes in each population, and MR reproduction. This audit does not identify a causal variant or establish mediation in thymus or orbit.

Colocalization hypotheses and regional inputs

coloc.abf received beta/varbeta data for disease (dataset 1, case-control type) and expression (dataset 2, quantitative type, sdY = 1). Outcome allele frequency was not used in the Bayes factors. Complete available cis-eQTL statistics, not just significant eQTLs, were matched by identifier and allele compatibility. H0 denotes neither trait associated; H1 disease only; H2 expression only; H3 both traits associated through distinct variants; H4 both traits associated through one shared variant. PP.Hk is the posterior probability of hypothesis k, conditional on the model and priors. Dataset order is essential to interpreting H1 and H2.

Primary priors were $p1 = p2 = 10^{-4}$ and $p12 = 10^{-5}$; sensitivity p12 values were 5×10^{-6} and 10^{-6}. The PP.H4 ≥ 0.80 threshold was a reporting convention, not a significance test. H0–H4 sum to one apart from rounding. Values displayed as <0.001 are nonzero full-precision probabilities in the workbook. The top_snp is conditional on H4, not necessarily the disease lead variant. H3 is a competing distinct-variant model, not a count of causal signals. Multi-signal colocalization was not performed.

Four frequency/availability scenarios were used: original_reference (original available reference frequencies), paired_reference and paired_eqtlgen (the same variants with the respective frequencies), and available_eqtlgen (all matching variants with valid cohort frequencies). There are 1,440 gene–outcome–scenario–prior rows. The additional TSHR analysis comprises four windows × five outcomes × four scenarios × three priors = 240 rows. Windows were the full source cis-region, ±500 kb and ±250 kb around the recorded GRCh37 GenePos anchor, and the five-outcome variant intersection. The full cis-source contained 7,309 eQTL rows; all three FinnGen full-region analyses matched 6,579 variants, and the five-outcome intersection contained 4,030. Raw outcome regional scans checked whether the strongest disease variants survived matching. New candidate coverage checks also examined marginal expression strength; the absent rs5759410 lead in BBJ was checked against variant aliases without finding a replacement.

For *TSHR*, *IGF1R* and *CTLA4*, colocalization and MR in FinnGen GD and strict ophthalmopathy were independently reproduced from re-downloaded FinnGen R12 regional statistics using two separately written implementations, with the original ophthalmopathy endpoint as a control. The allele-compatible variant sets were the same (6,675, 7,850 and 4,993 variants), and the top variant was

the same in all 27 gene–endpoint–prior settings, including the control. The reproduction used gnomAD v2.1.1 European frequencies, which were available for more of these variants than the 1000 Genomes European frequencies used in the reported analysis (6,674, 7,850 and 4,992 analysed, versus 6,579, 7,740 and 4,924). On the control endpoint, PP.H4 changed by at most 4.1×10^{-4}; for *TSHR* this difference came from the additional variants. Reported new-endpoint PP.H4 values were reproduced within 1.2×10^{-3}, and no setting changed side of 0.80. The *TSHR* strict-ophthalmopathy value at p12 = 10^{-6} was 0.449 rather than 0.451 (0.451 on the reported variant set), and *IGF1R* PP.H2 and PP.H3 differed by up to 0.003, as on the control endpoint. MR estimates were reproduced to numerical precision on the original exposure scale.

Frequency substitution and precision benchmarks

The published eQTLGen frequency release covered 26,609 participants, excluding Framingham. Reconstructing β_X and SE_X with fixed Z and N preserves their ratio up to floating-point precision. For a single-IV Wald estimate, frequency scaling changes the effect unit and SE together and preserves the normal-test P value; for IVW, variant weights can change. The new-endpoint MR dataset separates identical-variant scaling from repeat allele harmonization. *C4B* and *HLA-G* were audited with identical retained variants and outcome effects, so their differences arise from scaling and weighting rather than a changed variant set. Large frequency discrepancies are documented, not assumed to be resolved by correct identifier matching.

The detection benchmarks use $(z_{1-\alpha/2} + z_{0.80}) \times SE$ as the approximate 80% minimum detectable absolute log-OR. Alpha was 0.05/2,544 for BBJ and 0.05 for the original UKB/FinnGen comparisons. These are precision summaries using observed SEs, not prospective sample-size calculations, post hoc proof of power, or power of colocalization/the combined filter. The Detection_original worksheet retains these three-outcome results. Applying the same formula to the MR_primary standard errors for FinnGen GD at alpha 0.05/2,544 gave 24.8% and 52.2% of 2,480 estimable genes with 80% power for ORs of 1.5 and 2.0. At alpha 0.05, the detectable *IGF1R* ORs were 1.72 in BBJ, 1.39 in UKB and 1.36 in FinnGen GD.

Fixed-panel tissue extension

The post hoc tissue extension retained a frozen panel of *TSHR* and *IGF1R* in five standard-release GTEx v8 tissues. QTL sample counts were 574 (thyroid), 483 (cultured fibroblasts), 581 (subcutaneous adipose), 469 (visceral adipose) and 670 (whole blood). The original v8 archive supplied significant variant–gene pairs and eGenes records for all tested genes; gene-specific release thresholds exceeded 5×10^{-8} for every eligible combination, permitting complete selection at the fixed threshold. Absence from the tested-gene file was distinguished from a minimum eQTL P value above

the threshold. The standard release includes multiple ancestries; the overall donor cohort was 85.3% European American, not a tissue-specific percentage.

Variants were restricted to ±1 Mb of the source GENCODE v26 TSS, $P < 5\times10^{-8}$, F > 10 and MAF ≥ 0.01. GTEx variant IDs identify GRCh38 REF and ALT, with slope referring to ALT. Actual effect-allele frequencies were reconstructed from each selected eGene MAF and ref_factor, not by treating MAF as ALT frequency. GTEx mappings linked GRCh38 variants to rsIDs and GRCh37 reference coordinates. PLINK clumping used 503 European reference samples, $r^2 < 0.001$ and a 10-Mb window; an independent genotype-based calculation reproduced selection. One of 20 thyroid *IGF1R* candidates was absent from the reference, while its strongest candidate was retained. All selected variants were non-palindromic and matched exactly in the five disease datasets; no proxies or relaxed thresholds were used.

Each of the four estimable gene–tissue combinations retained one SNP. Wald estimates used β_Y/β_X and $SE_Y/|\beta_X|$ with normal two-sided *P* values and 95% CIs; independent calculations reproduced all 20 estimates. The UKB linear-to-log-odds conversion was applied once. ORs represent one unit of inverse-normal-transformed normalized expression; they are not directly comparable to an intervention or to the reconstructed eQTLGen scale. Primary adjusted $P = \min(1, 10P)$ for FinnGen GD; secondary adjusted $P = \min(1, 40P)$ for the other datasets. The 30 unestimable cells remained in the planned families. Single-instrument analyses cannot support heterogeneity, leave-one-out or MR-Egger diagnostics. Significant-only eQTL pairs were not used for colocalization; full regional data were not obtained from the requester-pays resource, and no tissue posterior was calculated. Tissue_extension in the single Supplementary Dataset retains all 50 cells, availability, selected and excluded candidates, harmonized alleles and source statistics.

Orbital expression and reproducibility

Four patients with inactive TED and one control contributed orbital RNA-sequencing data. Orbital adipose tissue (approximately 2 ml, inferolateral compartment) was obtained during orbital decompression for persistent proptosis, diplopia or exposure keratopathy from patients with a documented active phase (Clinical Activity Score ≥3/7) who had completed immunosuppressive treatment, had remained inactive (Clinical Activity Score <3/7) for at least 6 months and had been euthyroid for at least 3 months. The control tissue was the nasal (medial) fat pad of one person without thyroid or autoimmune disease who underwent cosmetic upper blepharoplasty. Samples were snap-frozen; RNA was extracted with the RNeasy Lipid Tissue Mini kit (QIAGEN), and paired-end libraries (200–400 bp inserts) were sequenced on an Illumina NovaSeq platform, with two technical-replicate libraries per donor. Reads were checked with FastQC 0.11.7, trimmed with Trimmomatic 0.38, aligned to GRCh38 with HISAT2 2.1.0 and quantified as TPM with StringTie 2.1.3b. Technical replicates were combined by biological sample. Figure S1 shows sample-level transcript abundance

and log2[(mean TED TPM + 0.01)/(control TPM + 0.01)]. One control cannot estimate control-group biological variance reliably, and no differential-expression *P* value was used. The dataset is descriptive and did not influence candidate selection.

The workbook preserves exported full-precision analytical outputs, not spreadsheet reimplementations of MR. Readme and Column_dictionary identify each sheet, its scope and its columns. Reproduction of a calculation validates implementation, not instrument validity, causal assumptions or clinical generalizability.

Table S1. *TSHR* regional and prior sensitivity

Dataset	Full cis SNPs	Primary PP.H4	Lower-prior PP.H4	PP.H4 range across windows
BBJ GD	4,336	0.951	0.661	0.951–0.951
FinnGen GD	6,579	0.264	0.035	0.264–0.264
UKB HT	5,667	0.226	0.028	0.226–0.226
FinnGen GO	6,579	0.986	0.875	0.986–0.987
FinnGen GO strict	6,579	0.891	0.451	0.891–0.905

Lower prior: p12 = 10^{-6}; p1 = p2 = 10^{-4} throughout. Window range uses the full source cis-region, ±500 kb, ±250 kb and five-outcome intersection at the primary prior. It is not a confidence interval. Full regional disease lead variants were retained. All 240 settings are in TSHR_windows.

Table S2. Frequency-sensitive exploratory candidates

Gene / dataset	Frequency	IVs	OR (95% CI)	*P* value	Q-test *P*
C4B / FinnGen GD	Reference	2	1.63 (1.31–2.03)	1.10×10^{-5}	7.79×10^{-7}
C4B / FinnGen GD	eQTLGen	2	1.75 (1.31–2.33)	1.48×10^{-4}	1.36×10^{-8}
C4B / FinnGen GO strict	Reference	2	1.68 (1.44–1.95)	2.74×10^{-11}	0.122
C4B / FinnGen GO strict	eQTLGen	2	1.80 (1.46–2.23)	6.17×10^{-8}	0.0587
HLA-G / FinnGen GD	Reference	16	0.86 (0.72–1.03)	0.107	9.98×10^{-24}
HLA-G / FinnGen GD	eQTLGen	16	0.60 (0.47–0.75)	1.72×10^{-5}	2.26×10^{-10}
HLA-G / FinnGen GO strict	Reference	16	0.91 (0.76–1.08)	0.267	0.0231
HLA-G / FinnGen GO strict	eQTLGen	16	0.70 (0.53–0.92)	0.0122	0.131

C4B and *HLA-G* frequency sensitivity. Variant sets and outcome effects were held fixed. Frequency changes rescaled expression estimates and IVW weights. Q tests heterogeneity, not directional pleiotropy; its *P* value is not a gene-association *P* value. All 72 omission estimates and instrument weights are provided in the workbook.

Table S3. Tissue-specific expression and FinnGen GD

Gene / tissue	N	IVs	OR (95% CI)	*P* value	Adjusted *P*
TSHR / Thyroid	574	0	No qualifying IV	—	—
IGF1R / Thyroid	574	1	0.76 (0.56–1.02)	0.0692	0.692
TSHR / Cultured fibroblasts	483	0	Not tested	—	—
IGF1R / Cultured fibroblasts	483	1	1.11 (0.89–1.38)	0.359	1.000
TSHR / Subcutaneous adipose	581	1	0.36 (0.31–0.43)	1.28×10^{-34}	1.28×10^{-33}
IGF1R / Subcutaneous adipose	581	0	No qualifying IV	—	—
TSHR / Visceral adipose	469	1	0.50 (0.43–0.58)	4.09×10^{-19}	4.09×10^{-18}
IGF1R / Visceral adipose	469	0	No qualifying IV	—	—
TSHR / Whole blood	670	0	No qualifying IV	—	—
IGF1R / Whole blood	670	0	No qualifying IV	—	—

Fixed-panel tissue extension with FinnGen GD as the primary disease dataset. IVs are retained instrumental variants. ORs are per unit of normalized, inverse-normal-transformed expression. Adjusted $P = \min(1, 10P)$; the ten-comparison denominator includes unestimable cells. Not tested means the gene was absent from the tissue eGenes file; no qualifying IV means the minimum eQTL *P* did not meet 5×10^{-8}. All four estimable combinations used one SNP (F 32.1–61.1), so each *P* value equals that variant's disease-association *P* value. The adipose *TSHR* instruments (rs12101255 and rs74064796, 50 bp apart in *TSHR* intron 1) are themselves blood *TSHR* eQTLs near rs179252 and therefore re-express the regional *TSHR* association rather than providing independent evidence. The 40 secondary comparisons use a separate denominator of 40 and are retained, with all source statistics, in Tissue_extension. Cultured fibroblasts and adipose tissues are not orbital tissue. Original detection benchmarks remain in Detection_original.

Figure S1

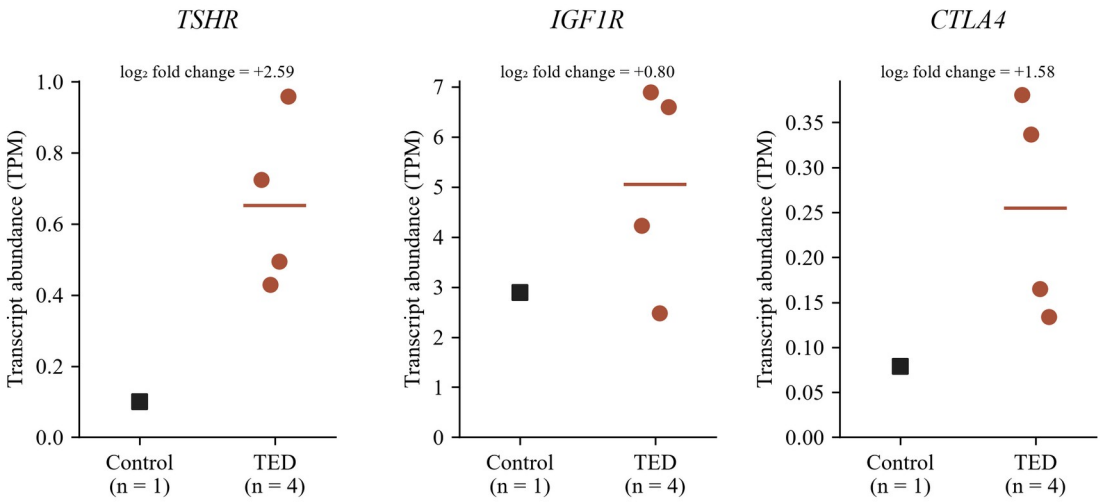


Figure S1. Descriptive orbital expression. Sample-level transcript abundance for *TSHR*, *IGF1R* and *CTLA4* in four TED patients and one control. Points are biological samples; horizontal lines show TED means. Fold changes use log2[(mean TED TPM + 0.01)/(control TPM + 0.01)]. Axes differ by gene. No inferential expression test was performed; these observations do not validate the genetic findings.

Figure S2

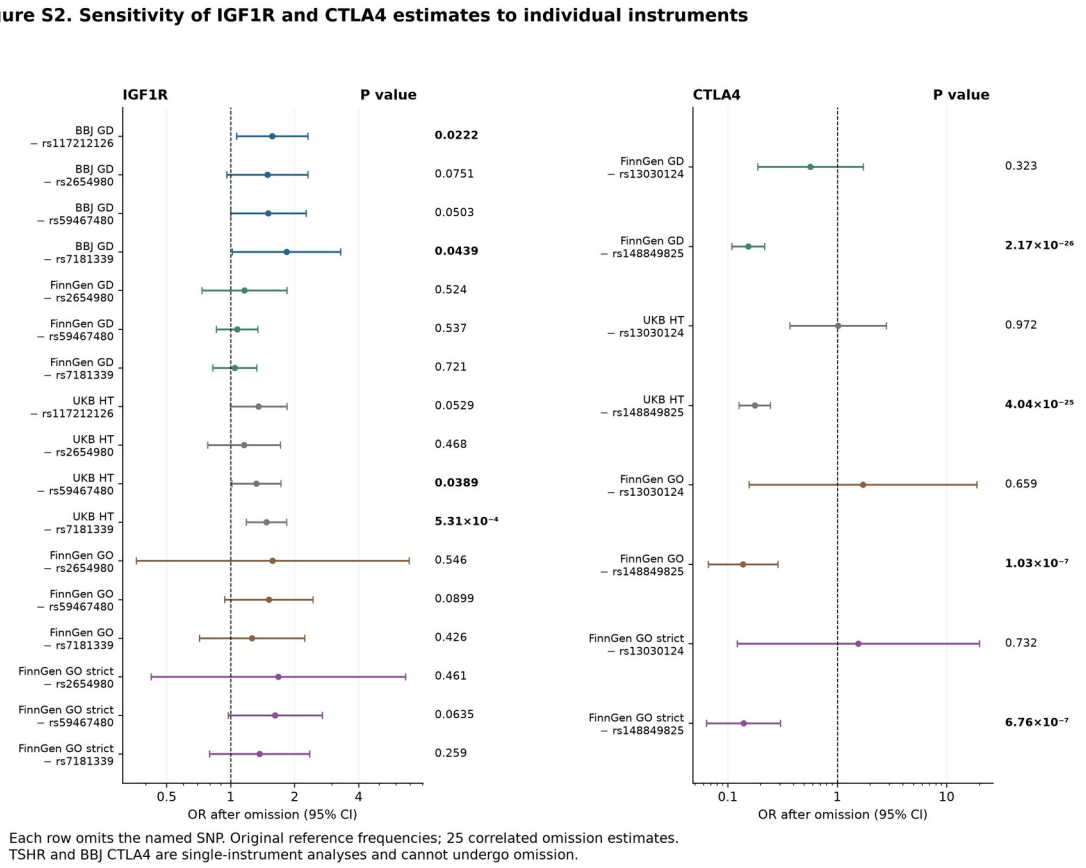


Figure S2. Instrument-omission sensitivity. Twenty-five reference-frequency SNP-omission estimates for *IGF1R* and *CTLA4* across the five datasets. The named SNP is omitted; Wald ratios apply when one remains and IVW when at least two remain. Horizontal lines show 95% CIs; the vertical reference is OR = 1. Bold *P* values mark nominal $P < 0.05$ only. Original full-set estimates are in Table 2. *TSHR* and BBJ *CTLA4* were ineligible because each has one instrument. These correlated post hoc diagnostics are not independent replication.